

Tarun Raheja

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EDUCATION

University of Pennsylvania

Master of Science in Scientific Computing; **GPA: 4.0/4.0**

Thesis Advisor : Prof Andrea Liu, **Topic** : Learning in Physical Systems

Philadelphia, USA

Aug. 2022 – Aug. 2024

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering (Hons.) in Computer Science; **GPA: 8.57/10**

Thesis advisor : Prof. Sreejith GJ, **Topic** : DMRG for Potts' Model

Hyderabad, India

Aug. 2015 – Aug. 2019

Graduated with First Degree Honours

PUBLICATIONS

1. Lambert, Neill* and **Raheja, Tarun*** and Cross, Simon and Menczel, Paul and Ahmed, Shahnawaz and Pitchford, Alexander and Burgarth, Daniel and Nori, Franco : QuTiP-BoFiN: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics. *Phys. Rev. Res.* 5, 013181 (2023).
2. **Raheja, Tarun*** and Pochhi, Nilay* : One Guardrail Is Not Enough: A Security Game for Deploying Complementary LLM Defenses. In *17th Conference on Game Theory and AI for Security (GameSec 2026)*. Talk (2026).
3. **Raheja, Tarun*** and Pochhi, Nilay* : One Crowd, Three Cascades: Information Diffusion in LLM-Agent Simulations Is Governed by Serialization, Not Society. In *COLM 2026 Second Workshop on Social Simulation with LLMs*. (2026).
4. **Raheja, Tarun*** and Pochhi, Nilay* : DimPFN: Units as an Exact Symmetry for In-Context Learning of Physical Laws. In *COLM 2026 Second Workshop on Language Models for Scientific Discovery*. (2026).
5. **Raheja, Tarun*** and Pochhi, Nilay* : One Posterior, Many Samplers: Open Problems in Sampling-Based Planning and Control. In *RSS 2026 Workshop on Sampling-Based Optimization in Robotics*. Oral Presentation (2026).
6. Varkey, Jyotsna and **Raheja, Tarun** : Curricula as Compression: An Information-Theoretic Account of Educational Mismatch and the AI Reversal. In *EHBEA 2026*. Poster (2026).
7. **Raheja, Tarun*** and Pochhi, Nilay* : From RLHF to Direct Alignment: A Theoretical Unification of Preference Learning for Large Language Models. *arXiv:2601.06108* (2026).
8. **Raheja, Tarun*** and Pochhi, Nilay* : An initial exploration of using Persistent Homology for Noise-Resilient Tactile Object Recognition. In *NeurIPS 2024 Workshop on Touch Processing: From Data to Knowledge*. Oral Presentation (2024).
9. **Raheja, Tarun*** and Sinha, Raunak and Deepak, Advit and Healy, Will and Srinivasa, Jayanth and Lee, Myungjin and Kompella, Ramana : Enhancing Large Language Models through Transforming Reasoning Problems into Classification Tasks. In *Proceedings of the 2024 ACL Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pp. 6007–6016. ELRA and ICCL, Torino, Italia (2024).
10. Haputhanthri, Udith and Storan, Liam and Jiang, Yiqi and **Raheja, Tarun** and Shai, Adam and Akengin, Orhun and Miolane, Nina and Schnitzer, Mark J and Dinc, Fatih and Tanaka, Hidenori : Understanding and controlling the geometry of memory organization in RNNs. *arXiv:2502.07256 (Under review)*
11. **Raheja, Tarun*** and Pochhi, Nilay* : The Alignment Trilemma: A Theoretical Perspective on Recursive Misalignment and Human-AI Adaptation Dynamics. In *ICLR 2025 Workshop on Bidirectional Human-AI Alignment*. (2025).
12. **Raheja, Tarun*** and Pochhi, Nilay* : On the Limitations of Neural Networks for Option Pricing: Analysis of Volatility Regime Sensitivity. In *ICLR 2025: I Can't Believe It's Not Better: Challenges in Applied Deep Learning*. (2024).
13. **Raheja, Tarun*** and Pochhi, Nilay* : DAGnosis: A Benchmark for Complex DAG-Based Multi-Step Reasoning in Large Language Models. (*Under review*)
14. **Raheja, Tarun*** and Pochhi, Nilay* : Foundation Models Meet Continual Learning: Recent Advances, Challenges, and Future Directions. In *NeurIPS 2024 Workshop on Scalable Continual Learning for Lifelong Foundation Models*. Accepted (Poster) (2024).
15. **Raheja, Tarun*** and Pochhi, Nilay and Curie, F. : Recent Advancements in LLM Red-Teaming: Techniques, Defenses, and Ethical Considerations. *arXiv:2410.09097* (2024).

* Denotes first authorship

PATENTS

1. Cache-Generated Frequently Asked Questions Page.

T. Raheja, R. Sinha, W. Healy, J. Srinivasa, A. Deepak, R. R. V. R. Kompella, A. Payani.
US Patent 12,572,536 (Granted Mar. 2026).

2. **Automatic retrieval augmented generation with expanding context.**
A. Deepak, W. Healy, **T. Raheja**, J. Srinivasa, R. R. V. R. Kompella, R. Sinha.
US Patent App. 18/442,996 (2025).
3. **Memory-based function calling for logical formulation and inference for large language models.**
T. Raheja, A. Deepak, W. Healy, J. Srinivasa, G. Liu, R. R. V. R. Kompella, et al.
US Patent App. 18/439,966 (2025).
4. **Performance monitoring, mitigation, and retraining of large language models.**
A. Deepak, R. Sinha, W. Healy, **T. Raheja**, J. Srinivasa, R. R. V. R. Kompella.
US Patent App. 18/440,710 (2025).

RELEVANT EXPERIENCE

Accel

*AI Resident (to **Shekhar Kirani**)*

Bangalore, India

Feb. 2026 – Present

- Advising on **AI investments** and helping **portfolio companies**.

Kipo AI

Head of ML, Founding Machine Learning Engineer

San Francisco, USA

May. 2024 – Jan. 2026

- Built AI for **U.S. government supply chain** use cases. Built and owned ML and MLOps infra as the **first ML hire**, and Employee No. **2**; later led the ML team as **Head of ML**.
- Architected and implemented a state-of-the-art scalable production **NLP** and **Vision** pipeline for understanding electronics datasheets. Outperforming competitors in **speed**, **cost**, and **data granularity**.
- Architected and implemented a state-of-the-art scalable production **agentic AI-search** pipeline for electronics components. This arbitrarily-fine-grained, domain-aware, **multi-modal**, **spec-accurate deep-research** is novel to the electronics industry.
- Architected and implemented tooling for **MLOps**, **DataOps**, and **training infra** for in-house vision and NLP models for several in-house tasks, specifically **YOLOv10** and **BERT** instances.
- Critically helped the company secure **2 major contracts**, directly worked on demos to clients, attended several sales **trade shows**. Mentored **2 interns**. Helped grow the team to 10 people, directly onboarded **2 engineers**.

Cisco Research

*Machine Learning Research Intern (with **Jayanth Srinivasa**)*

San Jose, USA

May. 2023 – Aug. 2023

- Led project on a **novel reasoning algorithm** for LLMs and an **LLM Gateway**. Resulted in **four patent submissions**.

Linguistic Data Consortium

*NLP Research Projects Programmer (with **Prof. Mark Liberman**)*

Philadelphia, USA

Sep. 2022 – Jan. 2024

- Worked on **forced alignment** and **generalized information representation** schemes for robust NLP models.

UPenn Department of Physics

*Graduate Research Assistant (with **Prof. Andrea Liu**, **Dr. Nachi Stern**)*

Philadelphia, USA

Jan. 2023 – May 2024

- Worked on algorithms for **learning in physical systems**. Uncovered novel **physical symmetry breaking** phenomena.

J.P. Morgan & Chase

*Quantitative Researcher (**QR Rates team**)*

Mumbai, India

Aug. 2021 – Aug. 2022

- Worked on developing mathematical pricing and volatility models for linear and nonlinear **rates derivatives** products.
- Worked on an OPA framework for **hedging residual Greeks**. Published docs for the **IR Future risk model**.
- Migrated internal **model usage restrictions**. Worked on two **high impact internal projects** reporting directly to MDs.

humit.app

Founding Backend Engineer

Bangalore, India

Feb. 2021 – Aug. 2021

- Built the social music sharing service, Humit. Designed and deployed several **core features** end-to-end including a **recommendation system**, and **async growth reporting services**.
- Wrote the algorithm for **WeirdSpotify**, which went **viral**.

RIKEN, Theoretical Quantum Physics Laboratory

*Research Assistant (with **Prof. Franco Nori**, **Dr. Neill Lambert**)*

Wako-shi, Japan

Oct. 2019 – Nov. 2020

- Developed an open source library to simulate **open quantum systems** using the **HEOM** formalism. Modelled the environment as a mixture of **fermionic and bosonic baths**. Studied **exceptional points** of large non-hermitian systems using a **novel block diagonalization** algorithm. Optimized **Noisy Mean Field Annealing** for NP hard problems using ideas from **quantum thermodynamics**.
- Worked on several research problems in **quantum information**, **open quantum systems**, and **combinatorial optimization**. Also worked on the **Coherent Ising Machine** project, and collaborated with **IBM Research Tokyo**.

- Worked on the admin team to **QuTiP**, an open-source package with **250k+** DAUs.

tensoralpha GmbH

Quantitative Developer Intern (with Dr. Thomas Schmelzer)

Berlin, Germany

Jul. 2020 – Aug. 2020

- Worked on statistical analysis of proprietary Crypto derivatives trading strategies. Researched a novel **tracker strategy**.

Microsoft Research

Research Intern (with Dr. Karthik Ramachandra)

Bangalore, India

Jul. 2019 – Sep. 2019

- Worked on **Project Froid**, on scalar **UDF inlining** for MSSQL. Improved if-else parsing from **exponential to linear**.
- Worked on **PyFroid**, a plugin to **Pandas**. Achieved **100x** speedup on certain operations through **lazy evaluation**.

IISER Pune

Undergraduate Thesis Researcher (with Prof. Sreejith GJ)

Pune, India

Jan. 2019 – May 2019

- Tailored the **Density Matrix Renormalization Group (DMRG)** algorithm to study **Potts' models**.

WorldQuant

Research Consultant (WebSim)

Remote

Dec. 2018 – 2019

- Offered consultant role for exceptional performance; built **high Sharpe, market-neutral alphas (Gold ranking)**.

Indian Statistical Institute (ISI) Calcutta

Cryptology Research Intern (ACM Summer School)

Kolkata, India

May 2018

- Selected among the **top 50 undergraduates** nationwide; studied **quantum and classical key exchange protocols**.

Zero Labs

R&D Intern (NLP)

India

May 2017

- Worked on **natural language processing** research and development.

SELECTED RESEARCH PROJECTS

Coherent Ising Machines (*NTT Research, 2020*) : Optical hardware for hard combinatorial optimization; each laser pulse in a fiber ring acts as a spin, coupled by measurement feedback.

Exceptional Points in Non-Hermitian Quantum Systems (*with Fabrizio Minganti, RIKEN*) : Studied exceptional points and exotic properties of large non-Hermitian systems using a novel algorithm.

Noisy Mean Field Annealing for Combinatorial Optimization (*with Neill Lambert, Clemens Gneiting, Ken Funo, RIKEN*) : Optimized NMFA for NP-hard problems using ideas from quantum thermodynamics.

Quantum Overlapping Tomography (*with Rudy Raymond, IBM Research; IBM Qiskit Asia Camp*) : Built an overlapping tomography module using perfect hash families; studied entanglement quality on IBM hardware.

PyFroid (*with Dr. Karthik Ramachandra, MSR India*) : A Pandas plugin achieving 100x speedups on certain operations through lazy evaluation.

Speech Event Segmentation (*with Neville Ryant, LDC*) : Metrics for speech segmentation via quantile analysis of intersyllable durations; stochastic point processes and neural equivalents for syllable event modeling.

Ontology-Aware Medical Concept Extraction (*with Prof. Lyle Ungar, Penn*) : BiLSTM-CRF and Transformers with UMLS embeddings for EHR named entity recognition.

Neural Network Potentials for Molecular Dynamics (*with Prof. Paris Perdikaris, Penn*) : ResNet-based NNPs for copper thermal expansion, validated with LAMMPS.

MPI Parallel Code Optimization (*with Prof. N. Anil, BITS Pilani*) : PETSc-based optimization for aerofoil sensitivity analysis on GPU-CPU hybrid systems.

Computational Geometry Library (*with Prof. Tathagata Ray, BITS Pilani*) : C++ implementations of convex hull algorithms, polygon triangulation, and point visibility.

Data Envelopment Analysis for Student Performance (*with Prof. Srinivasa Raju, BITS Pilani*) : DEA-based efficiency analysis of student input factors.

BLOGS & TALKS

Mastering the Art of Prompting for Large Language Models: Reducing Hallucination and Improving Reasoning : Cisco Outshift Blog (2024).

One Guardrail Is Not Enough: A Security Game for Deploying Complementary LLM Defenses : Talk, 17th Conference on Game Theory and AI for Security (GameSec 2026).

One Posterior, Many Samplers: Open Problems in Sampling-Based Planning and Control : Oral Presentation, RSS 2026 Workshop on Sampling-Based Optimization in Robotics.

An Initial Exploration of Using Persistent Homology for Noise-Resilient Tactile Object Recognition : Oral Presentation, NeurIPS 2024 Workshop on Touch Processing (Dec. 2024).

HONORS & AWARDS

Exa AI Discovery Agents Hackathon 2025 : Won first place among 50+ teams.
Sentient Foundation Werewolf AGIThon (AGI House) 2024: Won first place in agent strategy/jailbreaking hackathon.
CS50 Puzzle Day 2020 (Harvard x Facebook) : Scored 8/8. Global winner.
NTT Coherent Ising Machine (2020) : Participated in the NTT Coherent Ising Machine group's research.
IBM Qiskit Asia Camp 2019 : Invited by IBM.
ACM Summer School in Cryptology 2018 (ISI Calcutta) : Selected among the top 50 undergraduates nationwide.
WorldQuant Challenge : Achieved Gold ranking for high Sharpe, market-neutral, medium frequency alphas.
JEE Mains & Advanced : Ranked among top 0.3% nationwide, among 1.7 million candidates in the engineering entrance test.
Regional Mathematical Olympiad : Highest scorer in Bathinda region in India, for the first stage to the IMO.
Problem Solving Assessment Examination, CBSE : 99.8 percentile in national aptitude test, out of 1.5 million candidates.

ACADEMIC SERVICE, REVIEWING & MENTORING

Mentor, Conquest 2026 (BITS Pilani) : Asia's largest non-diluting startup accelerator
ICLR 2027
NeurIPS 2026
NeurIPS 2026 DynaFront Workshop
COLM 2026 LM4Sci Workshop
COLM 2026 Social Sim Workshop
ACM WebConf HCRS 2026
ICLR 2026
ICLR 2026 Workshops : I Can't Believe it's Not Better, MM Intelligence, P-AGI
ICLR 2025
NeurIPS 2025 Efficient Reasoning Workshop
NeurIPS 2025 Frontiers of Probabilistic Inference Workshop
NeurIPS 2024 RedTeaming GenAI Workshop
NeurIPS 2024 EvalEval Workshop
NeurIPS 2024 Workshop on Touch Processing

SKILLS

Programming : C, C++, Java, Python, MATLAB, R, SQL, $\text{\LaTeX}$, AWS, CI/CD
Technologies : Git, BASH, Django, Pandas, NumPy, PyTorch, TensorFlow, Spark, sklearn, Docker, Airflow
ML/SWE: Deep Learning, Computer Vision, Infrastructure, Data Structures, Algorithms, Scalable Design, Data Science